

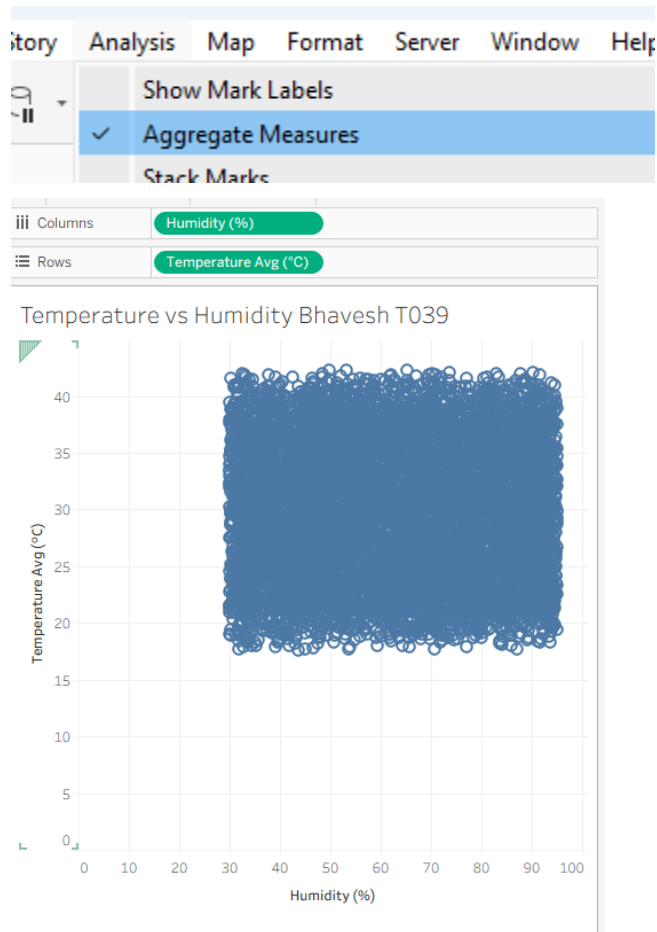
Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

Practical – VIII

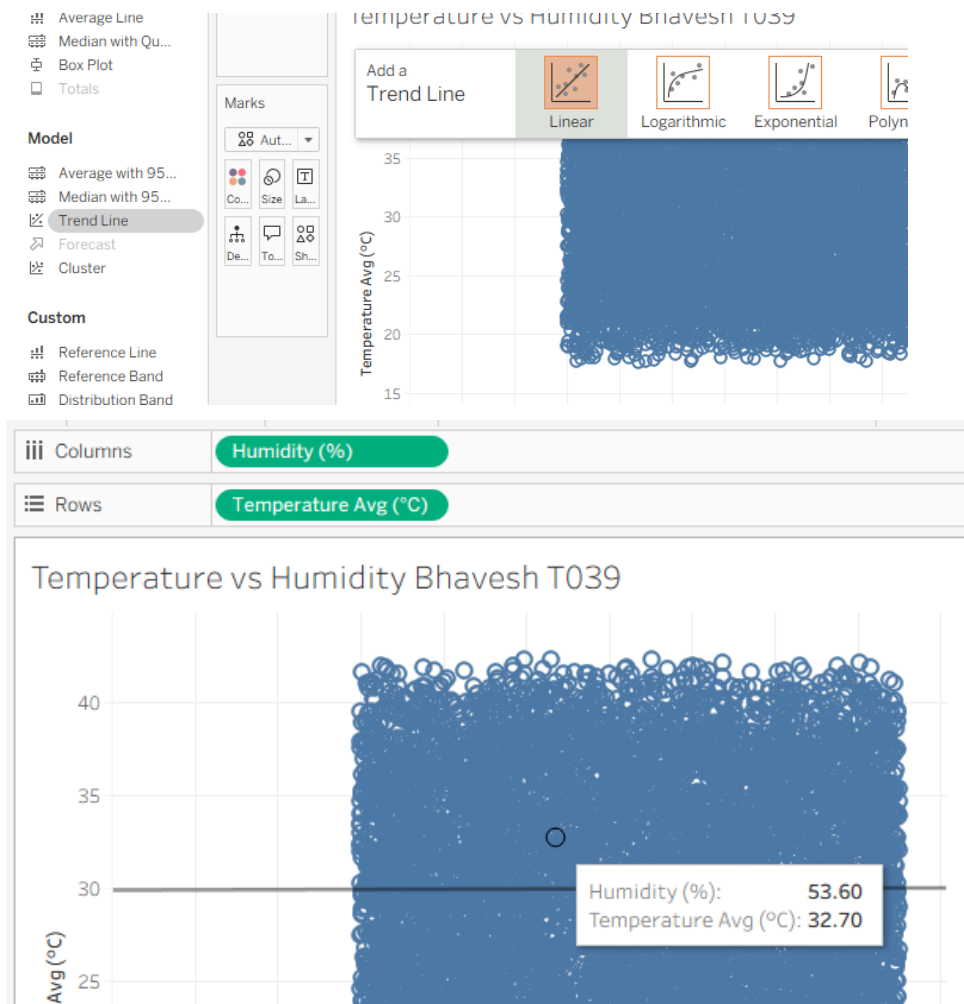
Roll No.:T039	Name:Bhavesh Ramakant Salaskar
Class:IT	Batch:1
Date of Assignment:27-08-2026	Date/Time of Submission:27-08-2026

Code:

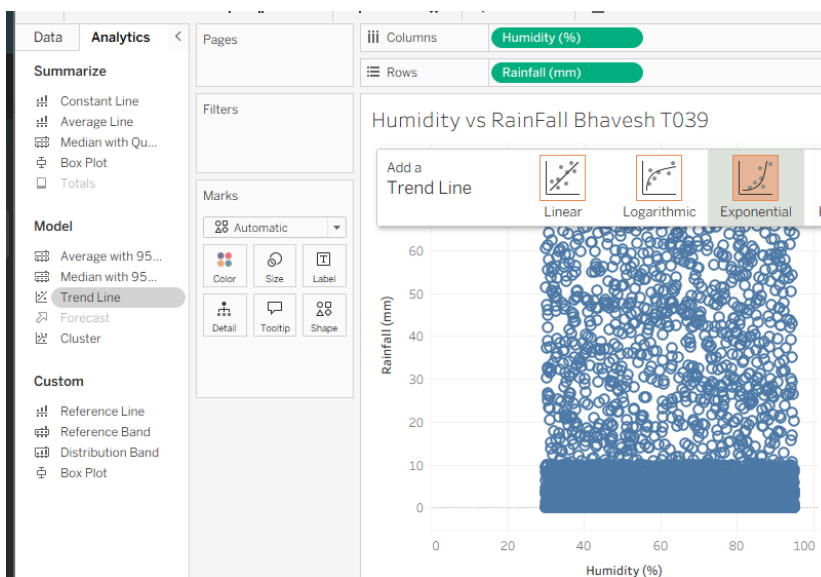
A: Add and Interpret Trend Lines (Linear, Exponential)

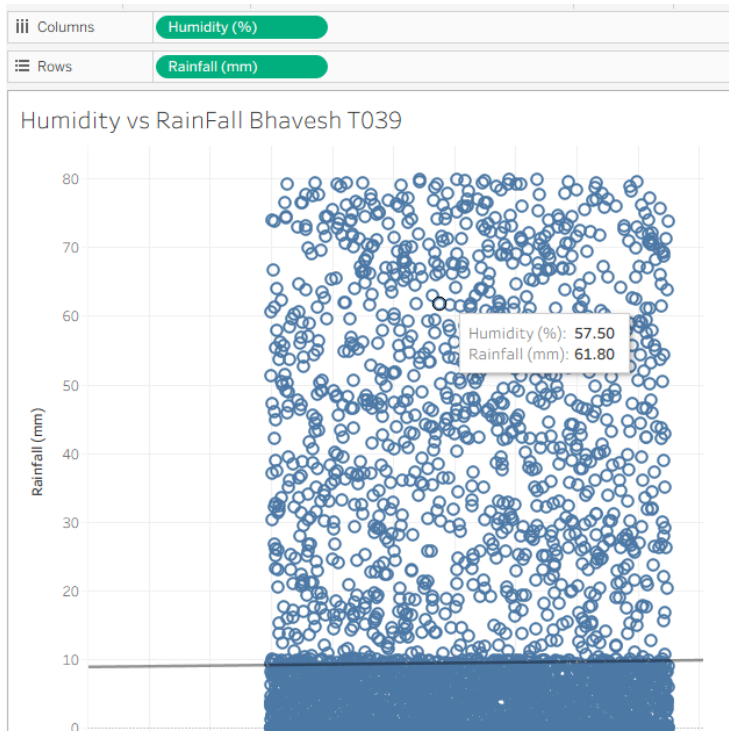


2. Add a Linear Trend Line

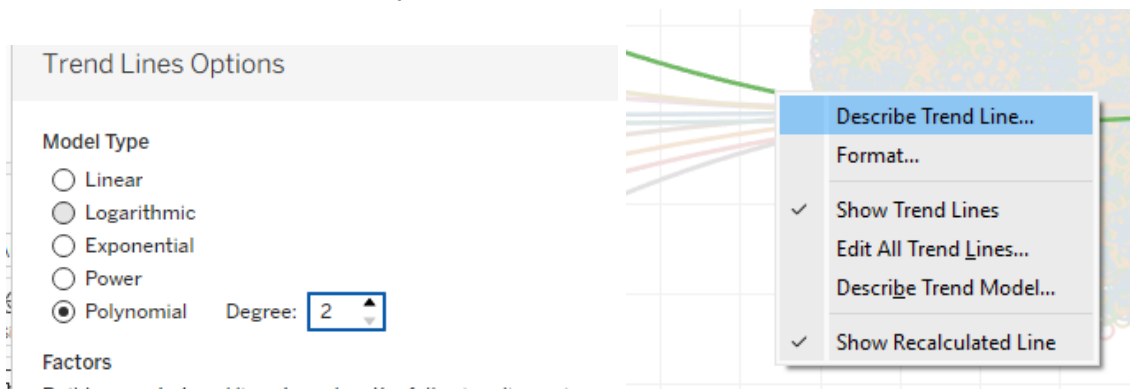


3. Add an Exponential Trend Line for Non-Linear Measures (Rainfall vs. Humidity)





B: Customize and Analyze Trend Models



Describe Trend Line

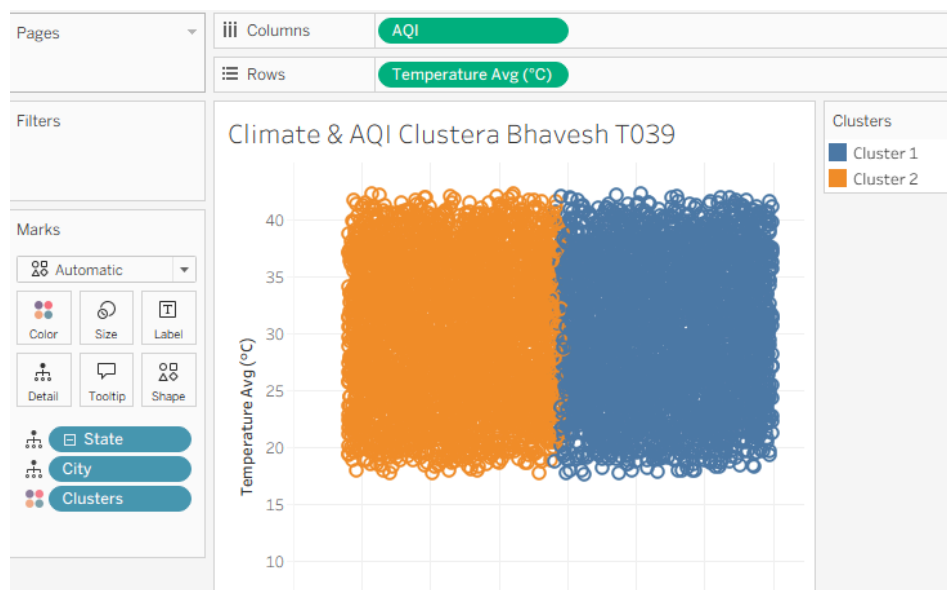
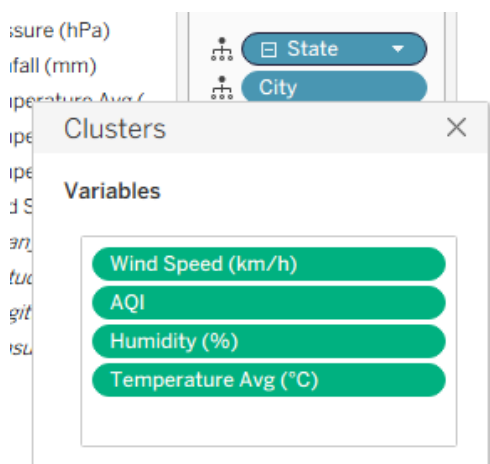
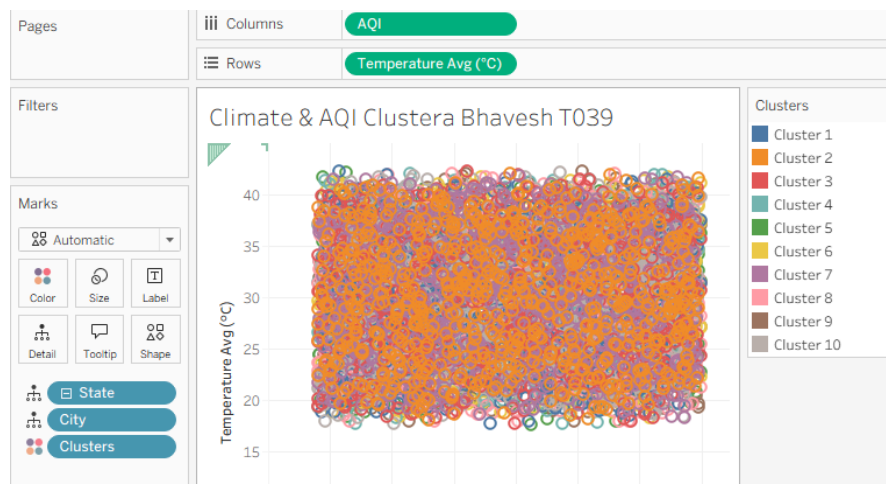
P-value: 0.127979

Equation: $\text{Temperature Avg (}^{\circ}\text{C)} = 0.00134169 * \text{Humidity (\%)}^2 + -0.158472 * \text{Humidity (\%)} + 34.2856$

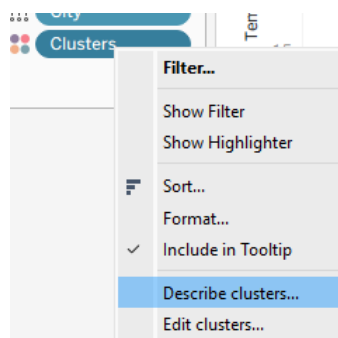
Coefficients

Term	Value	StdErr	t-value	p-value
Humidity (%)^2	0.0013417	0.0007287	1.84131	0.0659828
Humidity (%)	-0.158472	0.0924042	-1.71499	0.0867729
intercept	34.2856	2.74263	12.501	< 0.0001

C: Perform Clustering on Datasets



3. Analyze Cluster Profiles & Save as Dimension



Describe Clusters

Summary Models

Inputs for Clustering

Variables: Sum of AQI
Sum of Temperature Avg (°C)
Sum of Wind Speed (km/h)
Sum of Humidity (%)

Level of Detail: City, State

Scaling: Normalized

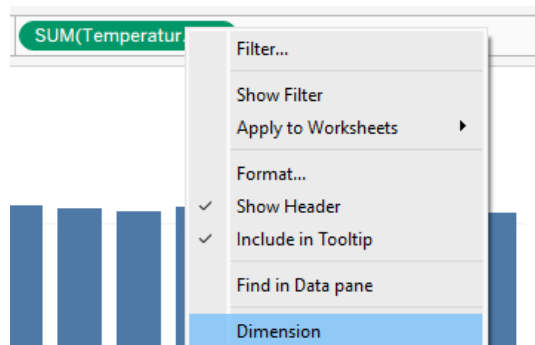
Summary Diagnostics

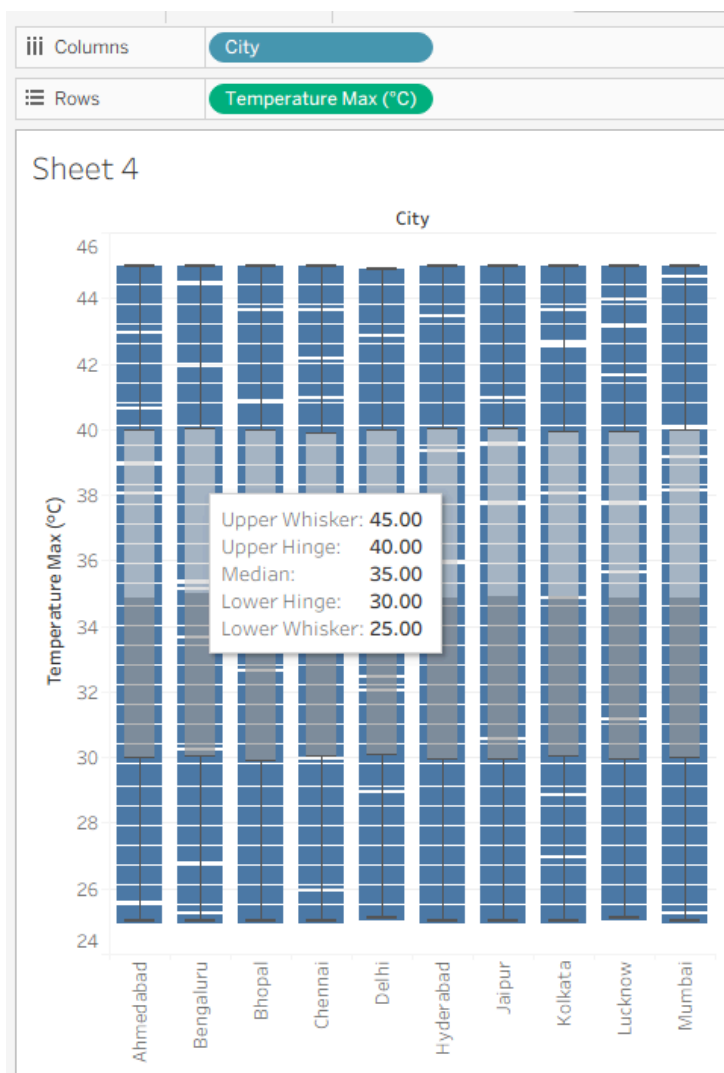
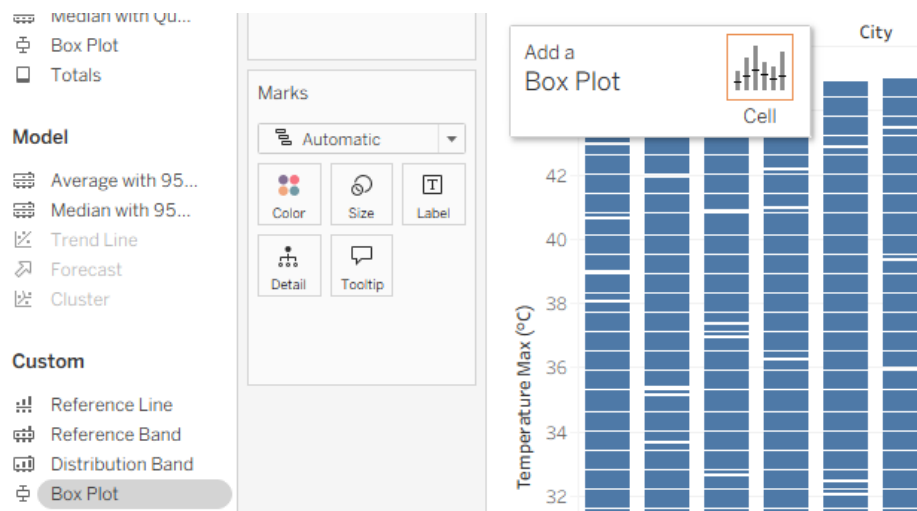
Number of Clusters: 2
Number of Points: 7310
Between-group Sum of Squares: 457.92
Within-group Sum of Squares: 1776.2
Total Sum of Squares: 2234.1

Centers

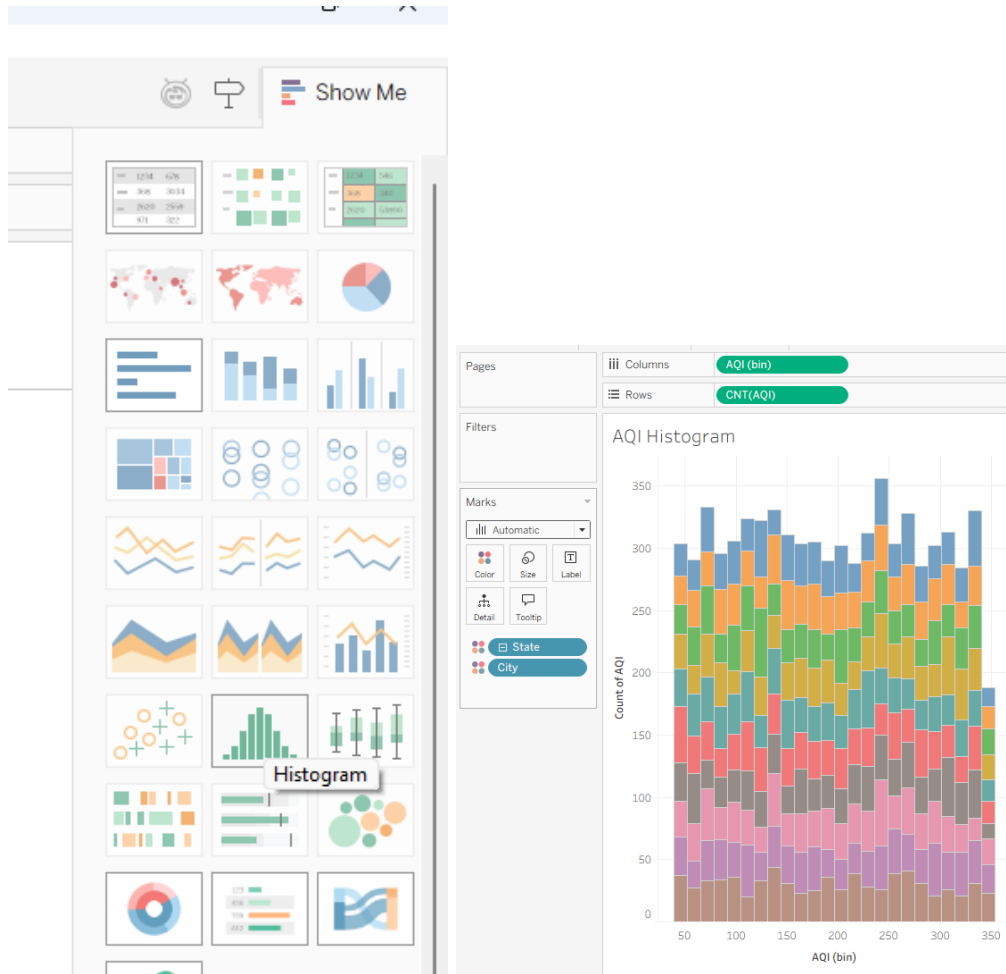
Clusters	Number of Items	Sum of AQI	Sum of Temperature Avg (°C)	Sum of Wind Speed (km/h)	Sum of Humidity (%)
Cluster 1	3657	271.03	29.999	13.412	62.375
Cluster 2	3653	116.41	29.962	13.634	62.932
Not Clustered	0				

D: Analyze Distributions using Built-in Tools

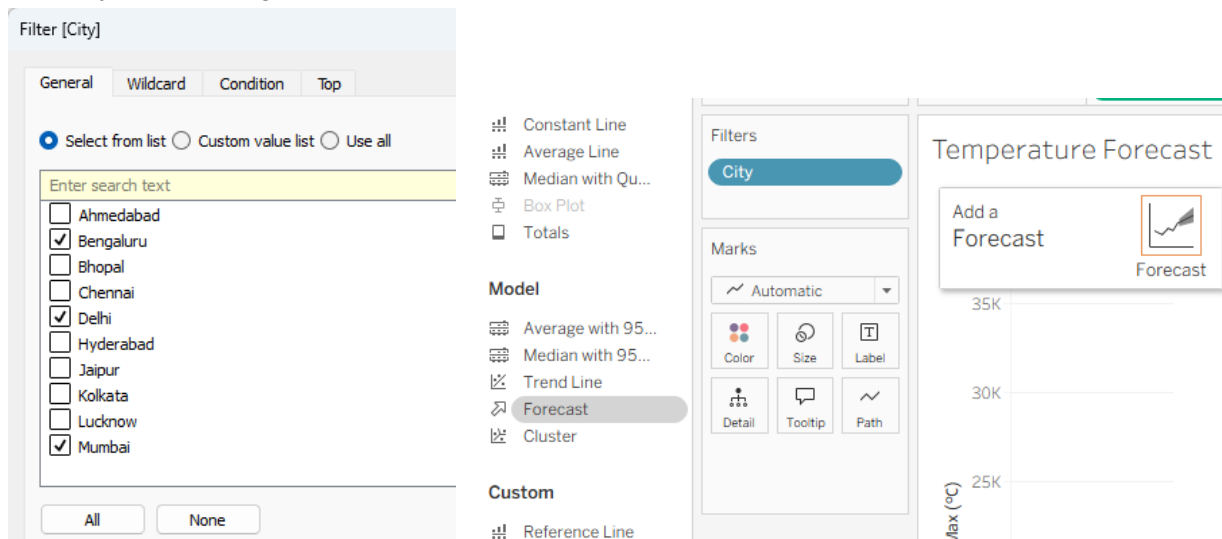


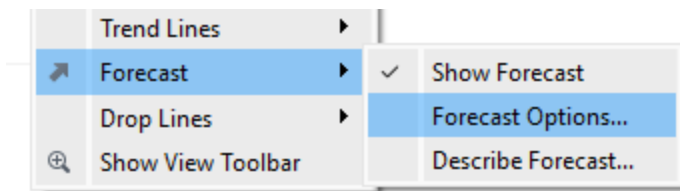


2. Build an AQI Histogram & Quantile/Percentile View



E: Apply Forecasting Techniques





Forecast Options

Forecast Length

☐ Automatic Next 13 months

☒ Exactly

☐ Until

Source Data

Aggregate by:

Ignore last: Months

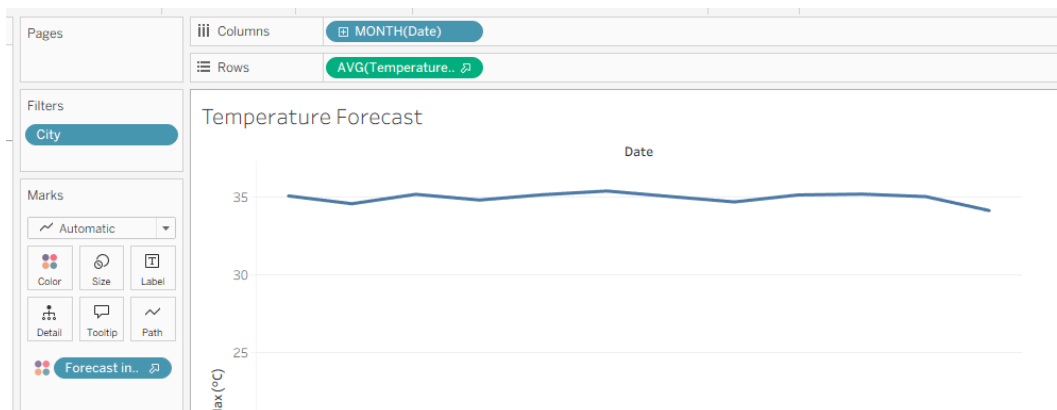
☐ Fill in missing values with zeroes

Forecast Model

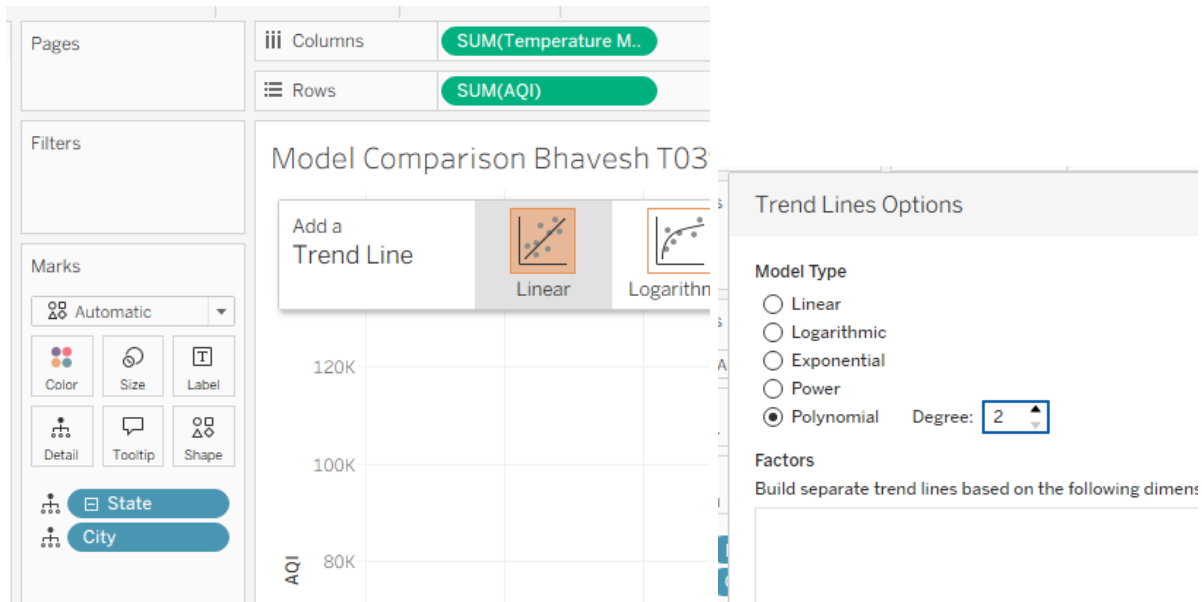
Automatically selects an exponential smoothing model for data that may have a trend and may have a seasonal pattern.

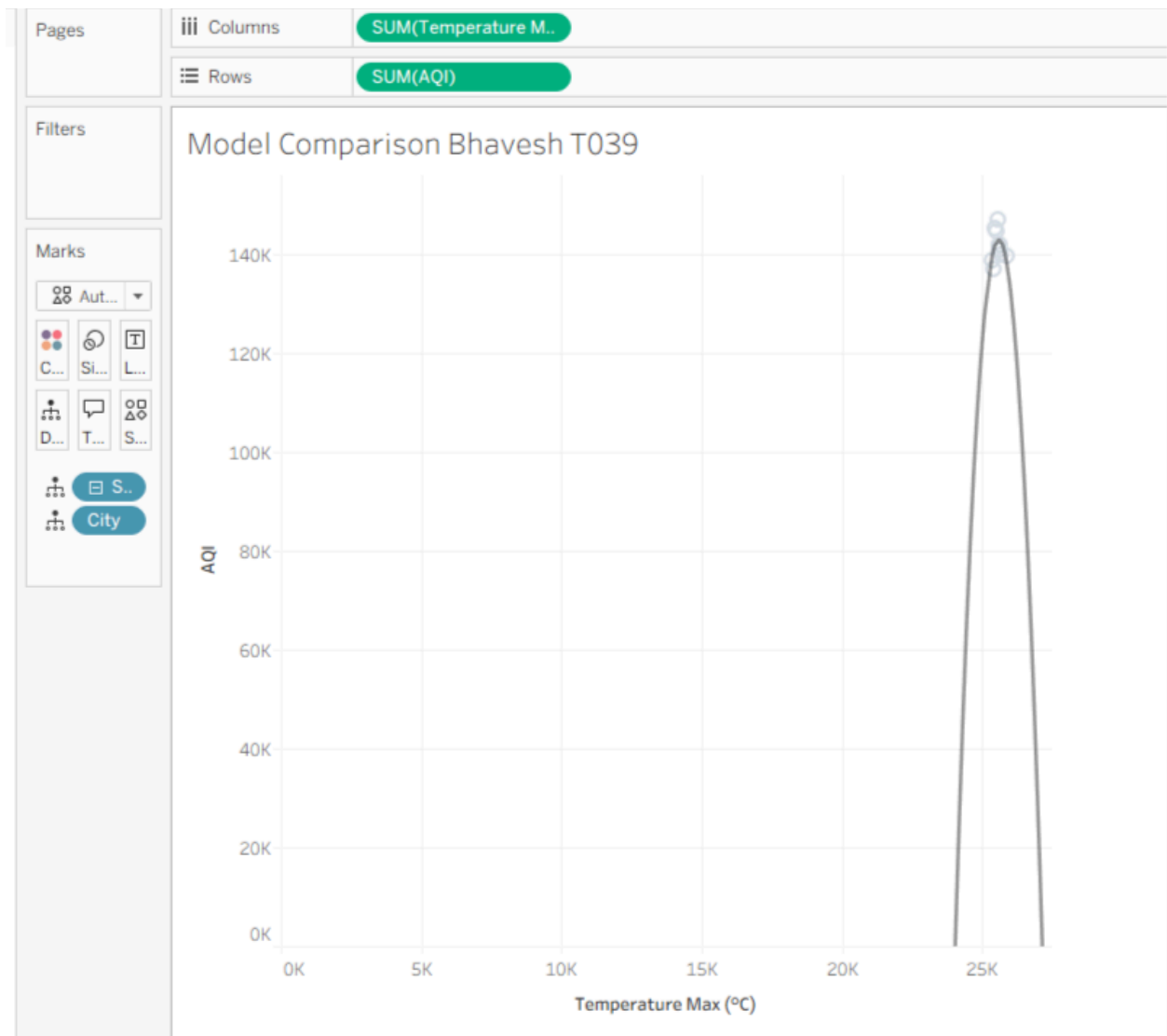
☒ Show prediction intervals

Currently using source data from January 2014 to November



F: Compare different Statistical Models





G: Export and Interpret Statistical Results

Describe Trend Line



P-value: 0.340466

Equation: $AQI = -0.0592329 \times \text{Temperature Max } (^{\circ}\text{C})^2 + 3030.3 \times \text{Temperature Max } (^{\circ}\text{C}) - 3.86139 \times 10^7$

Coefficients

<u>Term</u>	<u>Value</u>	<u>StdErr</u>	<u>t-value</u>	<u>p-value</u>
Temperature Max ($^{\circ}\text{C}$) ²	-0.0592329	0.0373375	-1.58642	0.156665
Temperature Max ($^{\circ}\text{C}$)	3030.3	1910.51	1.58612	0.156733
intercept	-3.86139e+07	2.44392e+07	-1.58	0.158118